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**Specific instructions for the Chemistry Contest**

Highly esteemed student:

- This exam was created by a Scientific Committee composed of former olympic students who work (mostly) at the University of Havana.
- **Complete your personal information.** Please verify with your ID card.
- The direction of the Provincial Education is responsible for ensuring that the exam is printed in a sufficiently high quality, allowing students to read the instructions, questions, figures, data, graphs, and tables. **No questions will be canceled due to poor printing.**
- Write all your information clearly in the space provided (first name, last name, school, municipality, province). **Do not write anything in the CODE field,** it is for the exclusive use of the coding manager.
- **Check that your exam is complete.** In the case of the exam on the first day, 10th grade should answer problems 1, 2, 3 and 4; 11th grade should answer problems 1, 2, 3 and 5; 12th grade should answer problems 1, 2, 5 and 6.
- All necessary constants, conversions, equations, and periodic table are included on the first page of the exam. This page will be removed during the coding process.
- At the beginning of each question, the grading is indicated, its value (out of a total of 100 points), as well as the value of each item (in marks).
- You can use a scientific calculator and a ruler.
- Please comply to the rules of approximate calculation. Attention! You have limited time and the exam has a large number of items. You do not have a space limit to answer, BUT please answer in an orderly manner, with appropriately sized and legible font, to facilitate the grading process.
- You must present all auxiliary calculations, justifying your reasoning and answers properly whenever necessary. **Unintelligible answers will not be graded.**
- Number the additional answer sheets at the bottom of each page and indicate the total number of pages, for example, “1 of 7,” “2 of 7,” and so on.
- If the exam instructor/proctor interferes with any of the above guidelines due to ignorance, exercise your right to protest and immediately request the intervention of the competent authority of the Ministry of Education, which is the institution responsible for administering the exam.
- The President of the Scientific Committee can be reached at +53-5045-4299.

You are a source of pride for Cuban education: Best of luck!

EXAM CODE (only for the use of the code manager)

Avogadro's Constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal Gas Law	$p \cdot V = n \cdot R \cdot T$
Gas constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$ $= 0,08205 \text{ atm} \cdot \text{L} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta U + W$
Zero in Celsius scale	273,15 K	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{c_{\text{red}}}{c_{\text{ox}}} \right)$
Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbance	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Vapor pression lowering	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Sphere volume	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Sphere area	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Circle area	$A_{\text{sp}} = \pi \cdot r^2$	Speed of light	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Length of a circumference	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Gravity force	$F_g = m \cdot g$	Gravity on Earth	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$ $1 \text{ bar} = 100 \text{ kPa}$ $1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$ $1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$ $1 \text{ dm}^3 = 1 \text{ L}$ $1 \text{ mL} = 1 \text{ cm}^3$ $1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																	2 He 4.003
2	3 Li 6.941	4 Be 9.012											5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305											13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	* 72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	** 104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]

*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]

EXAM CODE (only for the use of the code manager)

Problem 1. Hydrides (10% of total; 10th, 11th and 12th grades)

1.1	1.2	1.3	1.4	1.5	1.6	Total	Total
11	7	10	2	5	5	40	10%

One day, Lauren and Leticia were preparing a lesson about hydrides for tenth-grade students. Lauren wrote the following statement. Can you fill in the blanks to make it understandable?

1.1. Hydrogenated binary compounds can be classified into four main types. i) _____ hydrides are formed with the least electronegative metals. They have an _____ chemical bond and a large _____ partial charge on the hydrogen atom. They react vigorously with water, releasing _____, which reflects their potent _____ and _____ characteristics. ii) _____ hydrides are formed when hydrogen atoms occupy the empty spaces in the crystal lattices of the transition elements. They have chemical properties like the previous ones, although more attenuated, because in them the partial charge on the hydrogen atom is _____. They are electrical conductors in the solid state because the chemical bond they possess is _____ metallic _____. iii) _____ hydrides are formed with some chemical elements from groups 2, 11, 12, and 13, which have lower electronegativities and empty orbitals. The chemical bond in this case has intermediate characteristics. These hydrides react readily with the non-bonding electron pairs of Lewis _____. iv) Volatile hydrides are formed with non-metals and their acid-base behavior varies. Group 15 hydrides behave as bases because they have _____ in their molecular structure. For example, ammonia in aqueous solution increases the concentration of hydroxide ions in water. In contrast, group 16 and 17 hydrides in aqueous solution behave as acids because they have a large partial _____ on the hydrogen atom, thereby increasing the concentration of _____ ions in water. They react vigorously with alkali metals, releasing _____, which reflects their acidic and oxidizing characteristics. Hydrides of the first three types are _____ solids _____ at room temperature, while most type iv hydrides are _____, with _____ being a notable exception. Hydrogenated binary compounds of carbon, known as _____, can exist in all three aggregation states: as the chain length increases, the melting and boiling points _____ due to an _____ in _____ interactions.

For her part, Leticia found that ammonia has anomalous properties compared to the other hydrides in group 15, as shown in the following table.

Hydride	NH ₃	PH ₃	AsH ₃	SbH ₃	BiH ₃
Teb (°C)	-33	-88	-63	-17	17
α H-X-H (°)	107	93	92	92	91

1.2. Explain in detail why

- The boiling point of ammonia is unusually high.
- The bond angle is close to 90° in the hydrides mentioned above, except for ammonia.

Leticia and Lauren decided to demonstrate a practical application of a volatile hydride. Under a fume hood, they set up the equipment shown in the figure to generate a certain toxic and odorous gas *in situ*, which they then bubbled into 1,0 mol/L aqueous nitrate solutions.

The results obtained were as follows:

$\text{Al}(\text{NO}_3)_3$: No change was observed.

$\text{Fe}(\text{NO}_3)_3$: A black precipitate and a yellowish-white turbidity formed in the solution.

$\text{Co}(\text{NO}_3)_2$: A black precipitate formed and the pink solution became discolored.

$\text{Cu}(\text{NO}_3)_2$: A black precipitate formed and the blue solution became discolored.

$\text{Pb}(\text{NO}_3)_2$: A black precipitate formed.

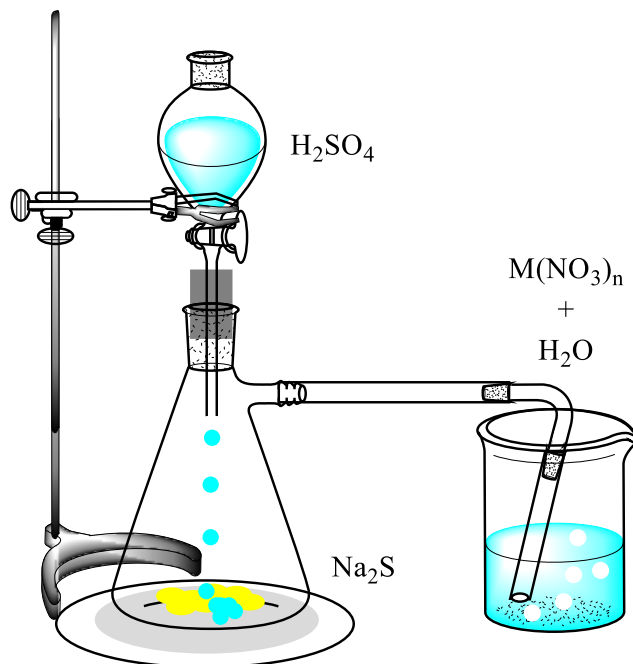
All the final solutions became acidic, although the first was the least acidic of all.

1.3. Formulate balanced chemical equations (ionic or molecular) corresponding to all reactions that occurred in the procedures described. Indicate the states of aggregation.

1.4. Why are Fe^{3+} , Co^{2+} and Cu^{2+} ions colored in aqueous solution?

1.5. Both contestants were struck by the fact that sulfide anions have a greater affinity for the cations Fe^{2+} , Co^{2+} , Cu^{2+} and Pb^{2+} than for the cations Al^{3+} and Fe^{3+} . Why?

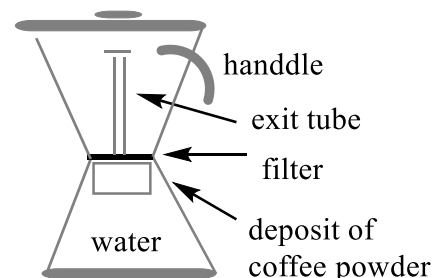
1.6. Identify the laboratory tools shown in the figure.



Problem 2. A can of coffee
(15% of total; 10th, 11th and 12th grades)

2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	Total	Total
2	2	5	3	18	6	2	10	12	60	15%

Most Cubans enjoy drinking coffee. If you visit a home or institution, you'll likely be offered a white porcelain cup containing about 50 mL of the beverage. Depending on the location, coffee can be heated with electricity, natural gas, coal, or firewood. Typically, coffee is prepared in a coffee pot, in an energy-intensive process. First, the water must be brought from room temperature, 25°C, to the boiling point, 100°C, consuming energy amount Q_1 . Then, a small amount of water turns into a gaseous state, causing an increase in pressure inside the lower reservoir and consuming energy amount Q_2 . Liquid water then mixes with coffee powder and extracts the main water-soluble component of coffee, caffeine; the formation of bubbles aids the extraction process. The excess pressure then causes the resulting solution to escape through the outlet tube. Finally, all that remains is to wait for the coffee to cool slightly and enjoy the beverage. A four-cup Cuban coffee pot can contain up to 200 mL of water; however, at the end of a "colada", only about 190 mL of coffee is obtained, which can be attributed primarily to evaporation losses.



To answer all the following questions, consider coffee to be an ideal dilute solution of the solute caffeine in the solvent water. Therefore, the heat capacity and density of coffee are equal to those of pure water, respectively, $c = 4,18 \text{ J}\cdot\text{K}^{-1}\cdot\text{g}^{-1}$ and $\rho = 1,00 \text{ g/mL}$. Also, use the thermodynamic data provided at the end of the question.

2.1. Coffee pots are made of metal, which facilitates both rapid heating and cooling. Why are metals good conductors of heat?

2.2. How much energy Q_1 (kJ) is required to heat all the water inside a full coffee pot?

2.3. How much energy Q_2 (kJ) is required to evaporate 10 mL of water?

2.4. What amount (mole) of water vapor exists inside the lower container when the last portion of the liquid is coming out? Assume that water vapor behaves in an ideal manner.

2.5. is the minimum amount of electricity (moles of electrons), natural gas (L at standard ambient conditions), coal (kg), and wood (kg) required to brew coffee in a coffee pot 1000 times?

In the village of Jatibonico, mornings are cold, typically around 15°C . Mario enjoys drinking coffee in the morning, but in his village, the beverage must be insulated in a thermos to keep it hot for a long time. The organizers gave all participants of the 56th ICHO a 1,0 L thermos, which Mario uses to store coffee. The heat flux Q/t can be calculated using the following equation: where k is the thermal conductivity coefficient of the material, A is the cross-sectional area through which heat energy transfer occurs, l is the thickness of the material, and ΔT is the temperature difference.

$$\frac{Q}{t} = \frac{k \cdot A \cdot \Delta T}{l}$$

2.6. Estimate the minimum time t (in min) required for all the freshly brewed coffee inside Mario's thermos to cool down to 15°C . Consider the cylindrical thermos (diameter 10,0 cm; height 14,0 cm) made of a material with $k = 0,15 \text{ W}\cdot\text{m}^{-1}\cdot^\circ\text{C}^{-1}$ and a thickness of 0,50 cm.

2.7. In practice, the cooling time is longer than you calculated. Why?

Emmanuel and *Riky* live in the city, but they enjoy hiking, biking, and overnight camping. Self-heating containers are more useful than thermoses. By pressing a button, they can break a film, separating a suitable solid chemical from the water in a small reservoir inside the container. An exothermic process then occurs, heating the entire container. The solid can be composed of a pure substance or a mixture, for example,

CaO , CaCl_2 , $\text{Mg} + \text{Fe}$, $\text{AlCl}_3 + \text{CaO}$, $\text{P}_2\text{O}_5 + \text{CaO}$, $\text{CuSO}_4 + \text{Zn}$.

2.8. Formulate the equations corresponding to the reactions that occur (if any) in each case. In the third formulation, the iron acts as a redox catalyst; it is not consumed in the process.

2.9. One of the products formed, calcium hydroxide, has a high enthalpy of formation, which makes the process exothermic. Estimate the value of this magnitude from the corresponding thermodynamic data. Use the Born-Haber cycle and the Kapustinskii equation to calculate the lattice energy.

Useful data:

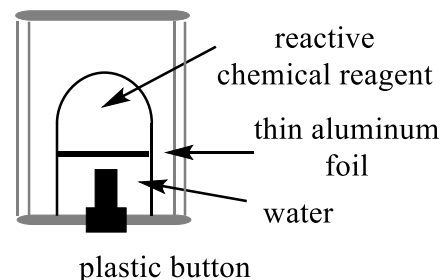
In Cuba, the electricity grid supplies direct current with a voltage of 110 V. Natural gas is composed mostly of methane. The average heat of combustion of wood is 18 kJ/g.

$\Delta H_f^\circ(X)/\text{kJmol}^{-1}$: $\text{H}_2\text{O, liquid} = -286$; $\text{H}_2\text{O, vapor} = -242$; $\text{CH}_4 = -79$; $\text{CO}_2 = -394$;

The charge of an electron is $e = 1,602 \cdot 10^{-19} \text{ C}$; one Faradio is 1 mol of electrons.

The fundamental equation of calorimetry is $Q = m \cdot c \cdot \Delta T$, where c is the heat capacity of the material.

The following values are expressed in [kJ/mol]: $I_1(\text{Ca}) = 590$, $I_2(\text{Ca}) = 1145$, $\Delta H_{\text{EA}}(\text{OH}) = -177$ (electron affinity of the hydroxyl radical), $E(\text{O}=\text{O}) = 495$, $E(\text{H}-\text{H}) = 434$, $E(\text{O}-\text{H}) = 465$, $\Delta H_{\text{sub}}(\text{Ca}) = 178$.



Kapustinskii's equation, U is the lattice enthalpy in kJ/mol, r and z are the radii in pm and the charges of the ions, respectively, $\sum v$ is the total number of ions in the empirical formula, $\sum v = 3$ for $\text{Ca}(\text{OH})_2$.

$$U \approx -1,20 \cdot 10^5 \cdot \frac{\sum v \cdot |z^+| |z^-|}{r^+ + r^-} \cdot \left(1 - \frac{34,5}{r^+ + r^-}\right)$$

The thermochemical radii of calcium and hydroxide ions are 114 and 133 pm, respectively.

Problem 3. Electronegativity (10% of total; only 10th and 11th grades)

3.1	3.2	3.3	3.4	3.5	3.6	3.7	3.8	3.9	3.10	3.11	3.12	Total	Total
2	4	2	4	2	2	3	6	2	6	3	4	40	10%

In the Pre-Selection, there are two students named Carlos, who come from different provinces: one from Sancti Spiritus and the other from Camagüey. They are friends and don't mind the historical rivalry between their teams. One day, they decided to study the theoretical foundations of chemical bonding together, but they stumbled upon the obscure concept of "electronegativity", and a heated debate ensued. Electronegativity occupies a central position in describing chemical bonding, and since it was postulated by Linus Pauling in 1932, it has been a fundamental tool for understanding charge transfer, bond polarity and strength, reactivity, and various chemical properties.

3.1. Write the concept of electronegativity.

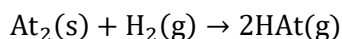
3.2. Explain how this property varies across the periodic table and the causes of the behavior described.

Although electronegativity χ cannot be measured, as it is not a physical quantity, it can be estimated indirectly. Carlos Daniel found the original definition used by Pauling, based on thermodynamic data of bond enthalpies (in kJ/mol) and taking hydrogen as an arbitrary reference, $\chi_{\text{H}} = 2,20$.

$$X_{\text{A}} - X_{\text{B}} = 0,102 \cdot \sqrt{\Delta H(\text{A} - \text{B}) - \frac{\Delta H(\text{A} - \text{A}) + \Delta H(\text{B} - \text{B})}{2}}$$

According to this definition, the additional stability of a covalent bond between two elements is due to the charge transfer or ionic contribution resulting from the difference in electronegativities.

3.3. Carlos Manuel found a periodic table and noticed that astatine has the same electronegativity as dihydrogen. What is the enthalpy change for the following reaction?



3.4. So, Carlos Daniel decided to calculate the electronegativity values of the remaining halogens to verify that they matched those in the table. What values did he obtain? $\Delta H^\circ(\text{A}-\text{B})/\text{kJ/mol}$: $\text{H}-\text{H} = 436$, $\text{F}-\text{F} = 156$, $\text{Cl}-\text{Cl} = 243$, $\text{Br}-\text{Br} = 193$, $\text{I}-\text{I} = 151$, $\text{H}-\text{F} = 569$, $\text{H}-\text{Cl} = 431$, $\text{H}-\text{Br} = 368$, $\text{H}-\text{I} = 297$.

3.5. The binding energy of fluorine is unusually low. What is the reason for this anomaly?

Carlos Manuel decided to apply the concept of electronegativity to explain the origin of bond polarity. When two atoms bond, an electron transfer occurs from the less electronegative atom to the more electronegative atom. Consequently, a partial charge δ , of equal magnitude but different signs, is generated on each atom, and the bond is said to be polar. The dipole moment μ of a molecule, which in the case of diatomic molecules, is a direct measure of bond polarity, can be experimentally determined using

Molecule	H-F	H-Cl	H-Br	H-I	H-At
μ (D)	1,86	1,11	0,79	0,38	~ 0

the formula $\mu = \delta \cdot d$, where d is the separation between the charges, in the case analyzed, the bond distance.

3.6. Explain the variation in the values that appear in the table above to Carlos Manuel.

3.7. Carlos Daniel discovered that during bond formation, the electronegativity of the atoms varies until the electronegativities of all the atoms are equal, according to Sanderson's Principle. Can you explain the origin of this variation and which atom increases/decreases electronegativity?

3.8. A Lewis acid is a substance with an empty orbital that can accept electrons, while a Lewis base is a substance with a filled orbital that can donate electrons. Help Carlos Manuel to identify which of the following pairs of substances are Lewis acids or bases, and specify which substance is the stronger acid or base for each pair; explain your answer.

i) $\text{H}_2\text{O}/\text{H}_3\text{N}$, ii) BCl_3/BI_3 , iii) $\text{HO}^-/\text{H}_2\text{O}$, iv) Cl_2/I_2

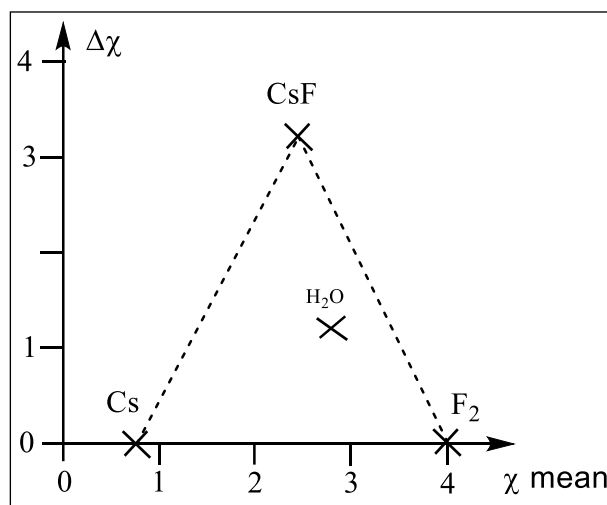
An oxide or hydroxide is more basic the greater the partial negative charge on the oxygen atom. Oxides or hydroxides are more acidic the greater the partial positive charge on the central and hydrogen atoms. The acid-base character of oxides and hydroxides varies regularly across the periodic table, depending on the oxidation number and the electronegativity of the atoms. Consider the element chromium, which forms a basic oxide, CrO , an amphoteric oxide, Cr_2O_3 , and an acidic oxide, CrO_3 .

3.9. Why does an increase in the oxidation number cause an increase in the acidity of the oxides?

With the data for each atom, Carlos Daniel decided to draw a Van Arkel-Ketelaar diagram, graphing the difference in electronegativity *versus* the average electronegativity for some substances.

atom	Cs	Li	B	H	C	N	O	F
χ	0,8	1,0	2,0	2,2	2,6	3,0	3,4	4,0

3.10. Find the coordinates of the substances B, BN, NF_3 , CH_4 , Li_2O and LiH , and place these substances in the appropriate places on the diagram. Indicate for each the predominant type of chemical bond (metallic, ionic, nonpolar covalent, polar covalent). Mark the exact position of each substance with an X and write its formula. Consider the examples provided in the figure (Cs, CsF , F_2 and H_2O).

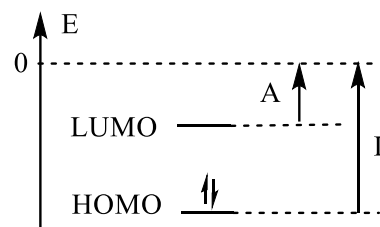


3.11. When Carlos Manuel looked at the graph, he immediately realized the relationship between the three extreme models of chemical bonding (ionic, covalent, and metallic) and electronegativity, and he took note of it. Can you complete the following statement?

When atoms of similar but _____ electronegativity combine, the resulting bond is metallic, for example, in metals and alloys. When atoms of _____ but _____ electronegativity combine, the bond is predominantly _____, as is the case in nonmetals and _____. On the other hand, atoms of very _____ electronegativity are held together by ionic bonding, as in salts.

At the end of the study session, students learned that it is also possible to calculate the electronegativity of a substance using the Mulliken method, from the electronic structure, and the electronaffinity **A** and ionization energy **I** data.

3.12. How must the values of **A** and **I** be like, for electronegativity to be high? Explain in each case.



Problem 4. Carbon (20% of total; only 10th grade)

4.1	4.2	4.3	4.4	4.5	4.6	4.7	4.8	4.9	Total	Total
8	14	6	3	5	4	10	10	20	80	20%

Carbon is a highly versatile and reactive chemical element, a constituent of essential compounds for life and industry. It exists in various allotropic forms, such as diamond, graphite, graphene, nanotubes, and fullerenes, of which graphite is the most stable. These allotropes can be considered to be formed by the repetition of two types of cyclic structural units of six atoms. In diamond, the structural unit is *chair*-shaped, containing only sp^3 -hybridized atoms, spaced 1,54 Å apart. The extension of the structural unit in space generates a three-dimensional network of tetrahedral atoms. In contrast, in graphite and graphene, the structural unit consists of a flat hexagon formed only by sp^2 -hybridized atoms, spaced 1,42 Å apart. The extension of the structural unit in space generates a sheet of only two dimensions (graphene), which can be stacked together to form a three-dimensional network (graphite), with a distance of 3,35 Å between the atoms of different sheets.

4.1. Draw the two structural units described above. Include only the six atoms that comprise them and all the bonds they form between them and other neighboring atoms.

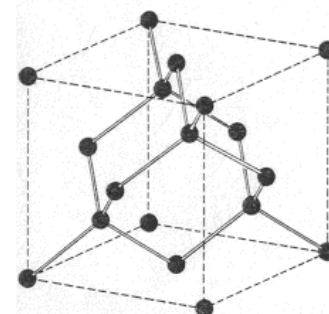
4.2. Explain in detail why

- Diamond is an electrical insulator, while graphite is a conductor.
- Diamond is very hard and is used in metal-cutting tools, while graphite is very soft and is used in powder form as a lubricant for heavy machinery.
- The six bond distances in the hexagonal structural unit of graphite are equal.
- Properties of graphite vary depending on the direction in which they are measured, a phenomenon known as anisotropy, while the properties of diamond are equal in any direction.

Osmani and Renny are very curious students, and during a bibliographic search, they discovered that carbon can also be present as diamagnetic diatomic molecules in the gaseous state.

4.3. Draw the molecular orbital energy diagram for the C_2 molecule. Note that the 2s and 2p orbital energies are similar, so s-p mixing exists.

Renny wants to study Designing, and because of this, he is deeply drawn to the study of the structure of substances. He knows that the regular structure of diamond can also be divided into small imaginary cubes with an edge length of 3,57 Å. The figure shows one of these cubes, also called a unit cell: there is a carbon atom at each vertex and in the center of each face, as well as an atom at four interior positions of the cube known as tetrahedral sites.



4.4. Explain why each unit cell can be considered to contain a total of eight carbon atoms.

4.5. Calculate the density of diamond in g/cm^3 . For reference, consider graphite as having an average density of 2,09-2,23 g/cm^3 .

In 1985, the family of carbon allotropes called fullerenes was discovered. Their three-dimensional structure is highly variable (spheres, ellipsoids, cylinders, and rings). The IUPAC defines fullerenes as “closed polyhedral cages formed entirely by n trivalent carbon atoms and having 12 pentagonal faces and $(n/2-10)$ hexagonal faces, where $n \geq 20$ ”. This discovery by Kroto, Curl, and Smalley earned them the Nobel Prize a few years later. The first member discovered was C_{60} , which is formed by 12 pentagons and 20 hexagons of sp^2 hybridized carbon atoms. It has a structure identical to a soccer ball and to the geodesic dome designed by the famous Brazilian architect, and for this reason, it is called buckminsterfullerene.

4.6. When Osmani asked Renny how many single and double bonds there are in C_{60} , it was easy for Renny to answer the question, since he was holding the soccer ball they were playing with; he just had to imagine the double bonds. Do you know the answer?

4.7. Noticing that Renny had cheated, Osmani got angry and decided to ask a more difficult question: When buckminsterfullerene reacts with oxygen, a small amount of fullerene oxide is produced. This oxide contains 97,83% carbon by mass and is very similar in structure to the starting material. What is the formula of this compound? How many double bonds and how many single bonds does it have?

Sometime later, the coordinator of the parents' council, Naylen, took the two students to a high-tech center at the Scientific Pole and, with the help of specialists, vaporized 10,0 g of graphite with a laser in an inert atmosphere. Analysis by HPLC chromatography and UV-Vis spectroscopy revealed the presence of the fullerenes C_{60} and C_{70} in a 3:1 molar ratio and a 20% total process yield.

4.8. Propose an expression to estimate the value of the combustion heat ΔH_{comb} of a fullerene C_n as a function of the enthalpy of combustion of carbon and other thermodynamic quantities that you consider appropriate. Why can $\Delta H_{\text{comb}}(C_n)$ be considered directly proportional to n ?

4.9. When the entire final mixture resulting from the previous process was combusted in an isothermal calorimeter, 334 kJ were released. Use this data and find the combustion enthalpies of fullerenes C_{60} and C_{70} .

Data: $1 \text{ \AA} = 100 \text{ pm} = 10^{-10} \text{ m}$ $d \text{ (pm): C-C} = 154, \text{C=C} = 133, \text{C}\equiv\text{C} = 120$ $\Delta H_{\text{comb}}^\circ (\text{C}) = -393,5 \text{ kJ/mol}$

ANSWER FROM HERE ON
(Except questions 1.1, 3.10 and 3.11)
